The Operator Mismatch: Non-Closure as a Formal Obstruction to Representation

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Abstract. General relativity is known to admit no non-trivial local, diffeomorphism-invariant observables — the *Problem of Observables* — while quantum theory is built entirely from operators on a Hilbert space. This asymmetry is usually treated as a technical obstruction to quantizing gravity by ordinary means, rather than as a fact with a structural origin. We propose a candidate origin, grounded in a specific algebraic decomposition: writing the sedenions as $\mathbb{S} = \mathbb{O} \oplus \mathbb{O} \cdot e_8$ via Cayley–Dickson doubling, we show that $\mathbb{O} \cdot e_8$ fails to close under the sedenion product — the product of two of its own elements lands entirely in $\mathbb{O}$ — and that this non-closure is, by itself, sufficient to formally prohibit any representation of $\mathbb{O} \cdot e_8$ in the ordinary sense. $\mathbb{O}$, by contrast, is genuinely representable: its commutator is proven elsewhere to be a non-Lie Malcev algebra, with a universal enveloping algebra whose central element is representable at every degree. We propose this asymmetry as a candidate, sufficient explanation for why gravity and quantum theory appear to "speak different languages" — not as a claim that this is the only possible explanation, which remains open.

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1. The Operator Mismatch

It is a well-established, if underappreciated, fact that general relativity does not possess non-trivial local diffeomorphism-invariant observables. For a local operator $\mathcal{O}(x)$, a diffeomorphism with parameter ξ^μ acts as $\delta_\xi \mathcal{O}(x) = \xi^\mu \partial_\mu \mathcal{O}(x)$, which vanishes only for constant $\mathcal{O}$; consequently, "no local operator is invariant under the gauge symmetry of gravity" (Khavkine, arXiv:1503.03754, and widely cited elsewhere as the *Problem of Observables*). What observables *do* exist in general relativity — Dirac observables, Kuchař's "perennials," Rovelli's relational observables built from a Material Reference System — are recovered only by coupling to matter and defining quantities relative to matter configurations; locality, in this framework, becomes "both relative and approximate." Gravity's own operator content, to the extent it exists at all, is borrowed.

Ordinary quantum theory has no such difficulty: local operators, acting on a Hilbert space, are its basic building blocks. This asymmetry — one side of physics genuinely operator-anchored, the other only ever relationally, approximately so — is the *Operator Mismatch*, and it is usually treated as a brute technical fact, or as evidence that the correct quantization procedure has not yet been found. We propose instead that it may have a precise, structural origin.

2. The sedenion split

Let $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}$ denote the sedenions, obtained from the octonions $\mathbb{O}$ by one further step of Cayley–Dickson doubling:

$$(a,b)(c,d) = (ac - \bar{d}b,\ da + b\bar{c}),$$

where $a, b, c, d \in \mathbb{O}$ and bar denotes octonion conjugation. Write $\mathbb{O}$ for the first copy $(a, 0)$ and $\mathbb{O} \cdot e_8$ for the second, $(0, b)$.

3. $\mathbb{O} \cdot e_8$ does not close

Proposition 1. For $m, n \in \mathbb{O} \cdot e_8$, the product mn lies entirely in $\mathbb{O}$.

Proof. Write $m = (0, b), n = (0, d)$. By the doubling formula, $mn = (0 \cdot 0 - \bar{d}b,\ d \cdot 0 + b \cdot \bar{0}) = (-\bar{d}b,\ 0)$, which has zero component in $\mathbb{O} \cdot e_8$. ■

This was independently confirmed by direct computation on a validated sedenion multiplication table (built from first principles via Cayley–Dickson doubling, checked against antisymmetry, $e_i^2 = -1$, octonion-block alternativity, and the presence of zero divisors before use): all 56 ordered pairs of distinct basis elements of $\mathbb{O} \cdot e_8$ produce results lying entirely within $\mathbb{O}$, without exception. A complementary check confirms $\mathbb{O}$ acting on $\mathbb{O} \cdot e_8$ *does* stay within $\mathbb{O} \cdot e_8$ — for $a \in \mathbb{O}, m \in \mathbb{O} \cdot e_8, am \in \mathbb{O} \cdot e_8$ and $ma \in \mathbb{O} \cdot e_8$ — so $\mathbb{O} \cdot e_8$ is a well-defined $\mathbb{O}$-bimodule. It is simply not closed under its *own* product.

This is a stronger and more basic fact than non-alternativity. Alternativity is a property of an algebra: a space already closed under its own product, whose associator merely fails a symmetry condition. $\mathbb{O} \cdot e_8$ has no self-product to be alternative or non-alternative *about* — Proposition 1 shows the product of two of its elements is simply not an element of it.

4. Non-closure is sufficient to prohibit representation

A representation of an algebraic structure A requires, at minimum, a linear map $\rho : A \to \mathrm{End}(V)$ satisfying some compatibility condition of the form $\rho(xy) = f(\rho(x), \rho(y))$ for $x, y \in A$. This expression presupposes $xy \in A$ — otherwise $\rho(xy)$ denotes the image of something outside the domain of ρ, and the expression is not merely false but ill-formed.

Proposition 2. No representation $\rho : \mathbb{O} \cdot e_8 \to \mathrm{End}(V)$ can be defined in the ordinary sense, treating $\mathbb{O} \cdot e_8$ as a self-standing algebraic object.

Proof. By Proposition 1, for $m, n \in \mathbb{O} \cdot e_8, mn \notin \mathbb{O} \cdot e_8$. The defining compatibility condition for a representation of $\mathbb{O} \cdot e_8$ requires $\rho(mn)$ to be defined in terms of $\rho(m), \rho(n)$; since mn does not lie in the domain of ρ, no such condition can be stated. This is a direct and immediate consequence of Proposition 1, stated separately to make its representation-theoretic content explicit. ■

This is offered as a *sufficient* obstruction: non-closure is enough, on its own, to rule out direct representation. We do not claim it is the *only* possible mechanism by which a structure could fail to admit operators; that stronger claim is not established here.

5. $\mathbb{O}$ remains genuinely representable

The mechanism of §§3–4 is general. It depends only on the Cayley–Dickson doubling formula itself, and applies identically at every level of the construction: the "second copy" produced by doubling any algebra in this family fails to close under its own product, by the same computation as Proposition 1, regardless of which algebra one starts from. The choice of $\mathbb{O}$ specifically — rather than, say, $\mathbb{H}$, whose commutator gives an ordinary, trivially representable Lie algebra, with no need for anything beyond standard enveloping-algebra theory — is not forced by this mechanism. It is motivated by results established elsewhere in this series: $\mathbb{O}$ is proven to be the unique point in the Hurwitz chain at which the commutator gives a genuinely non-Lie Malcev algebra that remains representable. The proposal of this paper depends only on some representable structure existing on one side of a doubling and a structurally non-closing one on the other; it does not depend on the representable side being Malcev specifically, only on Malcev being the richest such structure available at this particular, independently motivated level.

By contrast, $\mathrm{Im}(\mathbb{O})$ under the commutator is a genuine, non-Lie Malcev algebra — proven, elsewhere in this series, to be the unique such structure among the Hurwitz-chain algebras and their Cayley–Dickson continuations. Its universal nonassociative enveloping algebra rigorously exists (Pérez-Izquierdo–Shestakov, 2004), and for the specific octonion-derived algebraic R-flux construction studied in a companion paper, its central element is proven representable at *every* degree, by induction. $\mathbb{O}$, unlike $\mathbb{O} \cdot e_8$, is closed under its own bracket, and genuinely supports representation theory.

6. A floor beneath the wildness

The asymmetry of §§3–5 might suggest $\mathbb{S}$'s non-associativity is unbounded in principle. It is not. Every Cayley–Dickson algebra, at every level of doubling, remains **flexible** — $x(yx) = (xy)x$ — a classical result (Schafer). This was independently verified here: 300 random pairs drawn from the full 16-dimensional sedenion algebra satisfy flexibility exactly, without exception. Non-associativity in this family is real and, past the octonions, no longer alternative — but it is never total; flexibility is a structural floor that Cayley–Dickson doubling cannot break at any stage.

7. Proposal: a candidate resolution of the Operator Mismatch

If the physical distinction between what admits direct operators and what does not is correctly modeled by this algebraic split — $\mathbb{O}$ on one side, genuinely representable; $\mathbb{O} \cdot e_8$ on the other, structurally incapable of direct representation by Proposition 2 — then the Operator Mismatch of §1 is not a contingent gap awaiting a cleverer quantization scheme, but a formal consequence of examining two sides of one underlying structure that are not on equal algebraic footing. The established fact that gravity's observables are only ever relational, borrowed from matter (§1), would then have a precise counterpart here: $\mathbb{O} \cdot e_8$ is not inert — it acts as, and is acted on by, a genuine $\mathbb{O}$-bimodule, appearing directly in every product computed within $\mathbb{O}$ via the doubling formula's own cross-terms — but it never contributes an operator of its own, for the structural reason given in Proposition 2, not merely because none has yet been found.

This is proposed as a *sufficient* mechanism, and the pairing with physics is stated with its parts separated. Physics supplies exactly one sector that refuses local observables — gravity, the Problem of Observables of §1, its content read relationally through matter. Two candidate accounts of that refusal are compatible with the mismatch. On the first, developed constructively in this series' companion papers, gravitational quantities are *state-stratum* quantities: like temperature, entropy, and the modular flow, they are properties of the pairing between algebra and state rather than elements of the algebra — non-operators by design, relational among the representable side's operators exactly as position space itself is. This account requires nothing of $\mathbb{O} \cdot e_8$. On the second, $\mathbb{O} \cdot e_8$ is identified directly with the gravitational sector; it is retained as an open alternative, with the note that the constructive results of this series nowhere require it. On either account, two clarifications hold. The complement is neither external nor structureless: it is bound to $\mathbb{O}$ inside one algebra by the very multiplication that fails to close, and it is a faithful, irreducible Cayley bimodule over $\mathbb{O}$ (classified in the companion paper on the sedenion split), so that its structure consists entirely of its relations to $\mathbb{O}$ — what it lacks is not structure, and not relations, but representability. And none of this concerns space in its kinematic sense: positions, momenta, and rotations are operator-valued, and their algebraic home is necessarily the representable side. The sedenion level itself earns its place independently of any identification: it certifies the boundary — the specialness of $\mathbb{O}$ is defined by what fails one step above it — and it houses the three overlapping octonion subalgebras studied later in this series, each drawing half its directions from the complement. Nothing here is offered as a proof that this is the only structure capable of producing the mismatch; the pairing of the two sides with physics remains open.

8. Scope

This paper deliberately excludes the finer classification of $\mathbb{O} \cdot e_8$ as an $\mathbb{O}$-bimodule — in particular, whether it matches a named structure such as a Cayley bimodule in the sense of Jacobson (1954), and its relationship (if any) to the alternative-bimodule condition — since this classification is still under active investigation and not yet settled to the standard of the results presented here. It is deferred to a companion note.

References

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- Companion papers in this series: *The Imaginary Octonions Are the Unique Non-Lie Malcev Algebra in the Hurwitz Chain*; *The Central Element of Algebraic R-Flux*.

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